Abstract—Few-shot image generation remains challenging due to limited training data, which often leads to mode collapse and poor image quality. Traditional Generative Adversarial Network (GAN) inversion methods struggle to balance reconstruction fidelity with editability, particularly in attribute group editing frameworks where semantic consistency and fine-grained detail preservation are critical. The Stable Attribute Group Editing (SAGE) framework, while effective for few-shot scenarios, relies on the pixel2style2pixel (pSp) encoder, which produces entangled latent codes that compromise both reconstruction quality and downstream task performance. To address these limitations, this study proposes FeatureSAGE, an enhanced few-shot image generation framework that integrates the StyleFeatureEditor (SFE) encoder and Feature Editor module into the SAGE architecture. The SFE encoder jointly predicts both W^+ latent codes and intermediate feature tensors, enabling high-fidelity inversion while maintaining editability. Experimental evaluation on the FUNIT Animal Faces and 102 Flowers datasets at 1024×1024 resolution demonstrates that FeatureSAGE achieves substantial improvements over baseline methods. On the Flowers dataset, FeatureSAGE obtains a Fréchet Inception Distance (FID) of 45.60, representing 15.8% and 25.9% improvements over SAGE and AGE respectively, while maintaining perceptual diversity with an LPIPS of 0.4209. On the Animal Faces dataset, the framework achieves an FID of 49.60 (18.6% improvement over SAGE, 20.2% over AGE) with an LPIPS of 0.4811. These findings establish that feature-level editing combined with improved latent inversion can effectively address the fidelity-editability trade-off, advancing the state-of-the-art in attribute group editing for data-scarce generative modeling scenarios.

Index Terms—Few-shot image generation, Generative Adversarial Networks, StyleGAN2, GAN inversion, StyleFeatureEditor, attribute group editing, latent space, Fréchet Inception Distance, LPIPS, data augmentation

I. Introduction

A. Background of the Study

Generative Adversarial Networks (GANs) have become a foundational framework for high-fidelity image generation. In a GAN, a generator and a discriminator are trained in a minimax game: the generator learns to produce synthetic images that fool the discriminator into classifying them as real, while the discriminator learns to distinguish generated images from real ones [?]. This adversarial process has enabled GANs to achieve remarkable realism in generated images (e.g. faces, animals) across many domains. In particular, the StyleGAN family of architectures [?] introduced a novel style-based generator that has set new standards for image quality. StyleGAN maps an input latent code z through a learned mapping network into an intermediate latent w in a space called W. Each layer of the generator is then modulated by w via adaptive instance normalization (AdaIN), and stochastic noise inputs add fine detail. This style-based design yields an intermediate latent space W that is significantly more disentangled than the original input space Z. As a result, controlled manipulation of individual semantic attributes (e.g. hair color, pose) is possible by adjusting w or the per-layer style codes. Subsequent improvements (StyleGAN2, StyleGAN3) refined the architecture and training, further boosting image fidelity and easing inversion (mapping real images back to latent codes).

StyleGAN's latent representations come in several forms. The basic mapping from Z to W produces a single 512-D style code w per image. An extended latent space W^+ is also used, where a separate w vector is specified for each of the 18 synthesis layers. An even higher-dimensional feature space F can be formed by concatenating intermediate style and feature tensors (sometimes denoted F_k), capturing spatial details in the generator. Recent work has highlighted a trade-off across these spaces: moving from W to W^+ to F increases reconstruction fidelity but reduces editability [?] . In other words, encoding an image in the low-dimensional W space yields highly edit-friendly representations but may lose fine detail, whereas the high-dimensional W^+ or F spaces can reconstruct details at the cost of some entanglement and overfitting. StyleGAN2 in particular has shown that inversion in W^+ benefits from regularization (e.g. path-length regularization) to improve reconstructability and make projected images easier to attribute to latent codes [?].

To edit real images with a GAN, one must first perform GAN inversion: find a latent code that generates (nearly) the given image. This is a well-studied but challenging problem [?], [?]. Early inversion methods used optimization (iteratively updating z to minimize reconstruction error), which can achieve high fidelity but is slow and may produce out-of-distribution latents. Encoder-based methods train a neural network to predict the latent directly, enabling real-time inversion at the expense of some quality. For StyleGAN, [?], proposed Pixel2Style2Pixel (pSp): a feed-forward encoder that maps any input image to a sequence of style vectors in W^+ (one w per layer). The pSp encoder can invert faces (or other images) into W^+ accurately without test-time optimization, greatly simplifying downstream editing. Generally, as noted by [?], "inversion methods either directly optimize the latent vector to minimize the error for the given image". Encoder approaches like pSp offer fast inference, while optimization-based methods remain stronger on pure reconstruction quality. In either case, the goal of GAN inversion is to find an internal representation that contains as much image detail as possible and still allows semantic edits [?].

Once a real image is embedded in StyleGAN's latent space, attribute editing becomes possible. It has been observed that semantic attributes (e.g. "smile," "age," "eyeglasses") correspond to linear directions or subspaces in the latent space. For example, [?] showed that well-trained StyleGAN latent codes encode a disentangled representation under linear transformations. In practice, one can move the latent w of an image along a learned direction Δw to change a particular attribute (e.g. $w + \alpha\Delta w$) [?]. Many works have exploited this: e.g. Linear SVM or PCA over latent codes can find directions for "smile" vs "no smile," or for pose changes This allows controlled editing via vector arithmetic. Importantly, style-based latents like W and W^+ are far less entangled than raw Z, so edits tend to be more semantically meaningful [?], [?]. Still, some entanglement remains changing one attribute may inadvertently affect others. Recent methods (e.g. using subspace projection or network-based attribute classifiers) seek to further disentangle attributes. Moreover, higher-dimensional latent spaces (like W^+ or F) can encode finer identity details, enabling extremely high-quality reconstructions. [?] leverage this by introducing a StyleFeatureEditor: an encoder that predicts both a W^+ code w and an intermediate feature tensor F_k. By editing in both w and F_k, they reconstruct and preserve finer image details even under heavy edits. In summary, StyleGAN latent spaces provide a powerful substrate for attribute editing, but the choice of W vs W^+ vs F presents a trade-off between edit stability and detail, and sophisticated encoders are needed to balance both.

A parallel line of research addresses few-shot image generation and editing. Unlike standard GAN training (which requires large datasets), few-shot generation aims to adapt a model to a new category or attributes given only a handful of examples. Meta-learning approaches have been applied here: for instance, the FIGR model [?] meta-trains a GAN with the Reptile algorithm so it can generate new classes from as few as 4 examples. Other works like FUNIT [?] tackle few-shot domain translation, using image encoders to condition style generation on a few references. However, generating realistic variation from very limited data remains challenging. In particular, attribute group editing has emerged as a promising strategy for few-shot GANs. The idea is to collect many examples of known categories to learn generic attribute directions, then apply those directions to few-shot target categories. AGE learns a dictionary of ”category-irrelevant” attribute vectors (e.g. pose, color) from seen classes and ”category-relevant” class embeddings (mean latent per class) [?]. At inference, given one or a few examples of an unseen category, AGE embed them into W^+ (using pSp) and manipulates those latents by combining the learned attribute directions. Similarly, [?] propose a Transformer-style approach with modules for learning discrete attribute codebooks and prompt-driven editing to handle diverse unseen classes. These methods show that by factorizing and reusing attribute vectors, a pre-trained GAN can generate plausible images of new categories without full retraining [?]. Nonetheless, they often assume large amounts of labeled data for the seen categories and focus on one attribute at a time or on broad class attributes (like ”breed of dog” vs ”background color of bird”), rather than coordinated multi-attribute control.

Despite these advances, important gaps remain. Most GAN inversion and editing methods have focused on single-attribute manipulations or on scenarios with abundant data. Achieving stable, coordinated multi-attribute edits is still difficult, especially under few-shot conditions. For example, existing StyleGAN inversion methods either excel at reconstruction but produce entangled edits, or vice versa [?], [?]. Moreover, according to [?] , ”class-inconsistency” is a common-problem GAN-generated images for downstream classification. Class inconsistency happens when the generative model corrupts some minor but critical information versus real pictures of a certain category [?]. Few-shot models like AGE and TAGE demonstrate that group editing is possible, but they often do not explicitly ensure fidelity or identity preservation when combining many edits. Moreover, generation quality in few-shot setups can degrade due to mode collapse or identity drift. In face editing, for instance, we desire that an edited face remains clearly the same person (high identity similarity) while attributes like expression or hair style change. Conventional metrics like Fréchet Inception Distance (FID) and LPIPS capture distribution quality and perceptual similarity [?], [?].

We proposed that achieving improved and stable group-wise attribute editing under few-shot conditions required an integrated approach. FeatureSAGE is designed to address this by combining replacing the standard pSp encoder of SAGE with StyleGANEditor Inverter and Feature Editor. The integrated encoder (inspired by StyleFeatureEditor) jointly predicts a W^+ latent w and a feature tensor F, aiming for high-fidelity inversion as well as explicit control of feature-level detail. This architecture is intended to preserve identity and image realism (as measured by identity similarity and low FID/LPIPS) while achieving disentangled, controlled modifications of target attributes. In summary, the background works on GANs, StyleGAN latent editing, and few-shot group editing together motivate FeatureSAGE's goal: to provide stable, coordinated multi-attribute control in the latent space of a pre-trained GAN, even with only limited example images, without sacrificing image quality or attribute disentanglement.

B. Statement of the Problem

This study examines the main limitations of the current SAGE approach, specifically the use of the Pixel2Style2Pixel (pSp) encoder, which tends to yield latently entangled codes and diminished control over semantic edits. The limitations reduce the creation of semantic-consistent, high-fidelity images from few-shot instances. While GAN-produced images are visually realistic, loss of fine details from inversion heavily degrades downstream performance for classification. In approaches such as SAGE, using the reconstructed image for training lowers accuracy because of the mismatch between the latent embeddings and the original information, as well as the lack of fidelity in the reconstructions. The loss of information is carried over during the editing process, diminishing the performance of the generative augmentation. Enhanced inversion would increase both editability of the image and downstream task performance. Specifically, the study seeks to answer the following research questions:

1) Can integrating StyleFeatureEditor inverter and Feature Editor improve latent inversion fidelity and preserve image details for style-based editing within the SAGE framework?
2) Do these modifications influence the consistency, diversity, and quality of images generated in the few-shot setting, particularly for novel classes in datasets Flowers and Animal Faces?
3) How well does FeatureSAGE perform in terms of editing reliability and disentanglement in few-shot scenarios?
4) How does FeatureSAGE compare to existing editing based few shot image generation methods, AGE and SAGE, in terms of edit stability, identity preservation, and overall image quality?

C. Objectives of the Study

The objective of this study is to develop and evaluate FeatureSAGE, a reliable few-shot image generator framework that improves upon the SAGE framework by integrating a StyleFeatureEditor Inverter and Feature Editor

Specifically , this study's object is as follows: label=c.

1) Integrate StyleFeatureEditing to SAGE framework: integrate the inverter and feature editor pair to improve generated image quality. This aims to make a framework variant of SAGE that can be used for a more downstream (e.g. Data Augmentation) task via high reconstructability and editability without the usual tradeoff.
2) Evaluate reconstruction quality of the inverted images of Animal Faces and Flowers using FID and LPIPS.
3) Benchmark FeatureSAGE against baseline methods, specifically SAGE and AGE, to demonstrate improvements in edit stability, identity preservation, and overall image quality.

D. Significance of the Study

This study contributes to the few-shot image generation domain by enhancing the editability and interpretability of latent codes in the SAGE framework. By integrating the StyleFeatureEditor's encoder and feature editor, the proposed FeatureSAGE framework improves stability and consistency in attribute group manipulation under data-scarce cases which addresses the key limitation of the existing few-shot generative models.

Aside from methodological contributions, the proposed approach also has practical relevance and has several downstream applications. In fine-grained visual recognition, FeatureSAGE can generate identity-preserving variations of an animal or plant from a very few dataset, which helps train classifiers for species or breed identification in environments where collecting large dataset is impossible or difficult to acquire. In biometric work that focuses on identity, such as forensic image analysis and identity-based data augmentation, FeatureSAGE can generate multiple edited versions of an image while keeping the same person's identity. This makes the findings more realistic and more reliable. In medical and scientific imaging, researchers often have small datasets because privacy rules limit sharing, and gathering new images can be expensive. In some cases, the condition or disease is rare.FeatureSAGE generates realistic images that can serve as a synthetic dataset for training and validating machine learning models.

This study extends prior work on attribute group editing and GAN latent spaces, and it gives clear guidance for building data-efficient image generation systems that can support downstream applications that need consistent, reliable image synthesis.

E. Scope and Limitation

This study is limited to few-shot image generation tasks using the StyleGAN2 generator. It specifically focuses on improving the reconstruction and editability of the model by integrating the StyleFeatureEditing Encoder and utilizing the Feature Editing Module into the framework of SAGE.

The datasets used in this study are restricted to Animal Faces and Flowers which offer diverse but structured categories suitable for evaluating attribute preservation and class coherence. The study does not explore optimization-based inversion techniques or apply to GAN architectures other than StyleGAN2. Results and conclusions are therefore constrained to the SAGE pipeline with these enhancements.